National University of Computer and Emerging Sciences

FAST LAHOREFALL 2022

Assignment # 5 **Applied Physics** **Max. Marks: 50**

Section: BCS 1B

Instructions:

1. *First page of the assignment should be the page given below*
2. *Assignment should be hand-written. Typed assignment will not be accepted.*
3. *Use A4 Paper for the assignment. Do not use any other sheet*
4. *Please write the answers in the space given below.*
5. *Make diagrams wherever possible.*

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Question #1: During the 4.0 min a 5.0 A current is set up in a wire, how many (a) coulombs and (b) Electrons pass through any cross section across the wire's width?

(Marks 10)

Given:

Time $t = 4\,min \Rightarrow 4 \times 60 \Rightarrow 240\,sec$.

current $I = 5A$

Find:

$Q = ?$ (charge) $n = ?$ (No. of e^-).

Calculations:

$$Q = It = 5(240)$$

$$\boxed{Q = 1200\,C}$$

$$n = \frac{Q}{e} \Rightarrow \frac{1200}{1.6 \times 10^{-19}}$$

$$\boxed{n = 7.5 \times 10^{21} \text{ electrons}}$$

Question #2: A small but measurable current of 1.2×10^{-10}A exists in a copper wire whose diameter is 2.5 mm .The number of charge carriers per unit volume is $8.49\times10^{28}\ m^3$. Assuming the current is uniform, calculate the (a) current density and (b) electron drift speed

(Marks 10)

Given:

$I = 1.2\times10^{-10} A$

$d = 2.5\times10^{-3} m$

$n = 8.49\times10^{28} m^3$ (No. of e^- per unit vol)

Find: $J = ?$ (charge density) $v = ?$ (drift speed)

Calculations:

$J = \frac{I}{A} \Rightarrow \frac{I}{\pi r^2} \Rightarrow \frac{I4}{\pi d^2}$

$= \frac{(1.2\times10^{-10})4}{3.14(2.5\times10^{-3})^2}$

$\boxed{J = 2.4446\times10^{-5} Am^{-2}}$

$v = J/ne$

$= \frac{2.4446\times10^{-5}}{(8.49\times10^{28})(1.6\times10^{-19})}$

$\boxed{v = 1.79\times10^{-15} ms^{-1}}$

Question #3: A wire of Nichrome (a nickel–chromium–iron alloy commonly used in heating elements) is 1.0 m long and 1.0 mm^2 in cross-sectional area. It carries a current of 4.0 A when a 2.0 V potential difference is applied between its ends. Calculate the conductivity of Nichrome. **(Marks 10)**

Given:

$L = 1m$

$A = 1\times10^{-6} m^2$

$I = 4A$

$V = 2V$

To Find:

$\sigma = ?$ (conductivity)

Calculations:

conductivity $\propto \frac{1}{\text{resistivity}}$ $(\sigma \propto \frac{1}{\rho})$

so;

$\sigma \Rightarrow \frac{1}{\rho} \Rightarrow$ ~~[illegible]~~ $\Rightarrow \frac{1}{\frac{RA}{L}} \Rightarrow \frac{L}{RA} \Rightarrow \frac{L}{\frac{V}{I}A}$

$\sigma \Rightarrow \frac{LI}{VA} \Rightarrow \frac{1\times4}{2\times1\times10^{-6}} \Rightarrow 2\times10^{6}\Omega^{-1}m^{-1}$

$\boxed{\sigma = 2\times10^{6}\Omega^{-1}m^{-1}}$

Question #4: How long does it take electrons to get from a car battery to the starting motor? Assume the current is 300 A and the electrons travel through a copper wire with cross-sectional area 0.21 cm^2 and length 0.85 m. The number of charge carriers per unit volume is $8.49\times10^{28} m^{-3}$.

(Marks 10)

Given: $A = 0.21 \times 10^{-4} m^2$

$I = 300\ A$

$L = 0.85 m$

$n = 8.49 \times 10^{28} m^{-3}$ (No. of charge carrier per unit vol)

Find: $t = ?$

Calculations:

$V = \frac{J}{ne} \Rightarrow \frac{I}{Ane}$

$\therefore L = Vt$

$t = \frac{L}{V} \Rightarrow \frac{L}{\frac{I}{Ane}}$

$t = \frac{A n e . L}{I}$

$t = \frac{0.21 \times 10^{-4} \times 8.49 \times 10^{28} \times 1.6 \times 10^{-19} \times 0.85}{300\ A}$

$\boxed{t = 8.08 \times 10^{2} s}$ OR $\boxed{t = 8.1 \times 10^{2} s}$

Question #5: A potential difference of 3.00 nV is set up across a 2.00 cm length of copper wire that has a radius of 2.00 mm. How much charge drifts through a cross section in 3.00 ms?

(Marks 10)

Given: $V = 3 \times 10^{-9} V$

$L = 2 \times 10^{-2} m$

$r = 2 \times 10^{-3} m$

$t = 3 \times 10^{-3} s$

Find: $Q = ?$

Calculations: $Q = I \times t \longrightarrow$ (ii)

For I: $I = \frac{V}{R} \longrightarrow$ (i)

For R: $R = \rho \frac{L}{A} \Rightarrow \rho \frac{L}{\pi r^2} \Rightarrow \frac{(1.69 \times 10^{-8})(2 \times 10^{-2})}{\pi (2 \times 10^{-3})^2}$

$\therefore R = 2.69 \times 10^{-5} \Omega$ put in (i)

$\therefore I = \frac{3 \times 10^{-9}}{2.69 \times 10^{-5}} \Rightarrow 1.115 \times 10^{-4} A$

put 'I' ~~and 'R'~~ in equation (ii)

$\therefore Q = It = 1.115 \times 10^{-4} (3 \times 10^{-3})$

$\boxed{Q = 3.35 \times 10^{-7} C}$